

INTRO TO GITHUB

A hands-on workshop for your first collaborative GitHub workflow

repository → branch → commit → push → pull request



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After you check in:

Make sure to have

- VS Code or another IDE for our hands – on challenge
- A GitHub account
- Git downloaded on your computer

Check In



Today's goals:

Leave with a workflow you can repeat on your own projects.

1. Understand

- Git vs. GitHub
- Repositories
- Branches
- Commits
- Pull requests

2. Do

- Fork a repository
- Clone it locally
- Create a branch
- Change a file
- Commit + push

3. Collaborate

- Open a pull request
- Review a diff
- Merge a change

4. Practice

Complete your first contribution to a shared GitHub repository.

Ask questions while we demo. There are no “stupid” Git questions!

Git vs. GitHub

GIT

- Version control software
- Runs on your computer
- Tracks changes and project history
- Works offline
- Commonly used through the command line



Git = the tool

GITHUB

- Platform for hosting Git repositories
- Stores your repositories online
- Enables collaboration and code review
- Provides features like pull requests, Issues, and Actions



GitHub = the place + collaboration layer

What is a repository?

- A project tracked by Git
- Contains code, images, documentation, configuration, and Git history
- On GitHub, a repository gives collaborators a shared home for the project
- Example: a website repo can contain frontend/, backend/, README.md, package.json, etc.

Sample repository

```
my-project/  
├── src/  
├── public/  
├── README.md  
├── package.json  
└── .gitignore
```

The GitHub workflow we'll practice

1

Fork



Create your own copy of the workshop repo

2

Clone



Get your repo onto your computer

3

Branch



Create a safe workspace for your changes

4

Edit



Make your change

5

Commit



Save a snapshot of your changes

6

Push



Send your branch to GitHub

7

Pull Request



Propose your changes for review

8

Merge



Integrate into main

DEMO 1

Explore a GitHub Repository

What does a repository actually look like?

Clone: Get a GitHub repo onto your computer

Clone = make a local copy of the repository, including its Git history.

```
git clone https://github.com/USERNAME/REPO.git  
cd REPO
```

After cloning:

- You have a local copy of the project
- Edit locally in VS Code or another code editor
- Git tracks your changes locally
- Your local repository is connected to the GitHub repository

Branches: Work without breaking main

A branch is a separate line of development.

```
git switch -c add-about-page  
# make your changes  
  
git switch main
```

Mental model

main = stable project

feature branch = your workspace

You can experiment, commit, and review before merging.

Why use branches?

Keep unfinished changes away from the stable project.

Commits: Snapshots, not save buttons

A commit records a set of changes with a message and points to a specific state of the project.

```
git status
```

```
git add .
```

```
git commit -m "Add WiCSE  
about section"
```

Good commit message

Add WiCSE about section

Update navbar links

Fix mobile layout

Bad commit message

Fixed it lol

fix for this page

Any vague message with no clear explanation of fix

A good commit message tells you what changed.

Push: Send your commits to GitHub

- Commit = save your changes locally
- Push = send your commits to GitHub

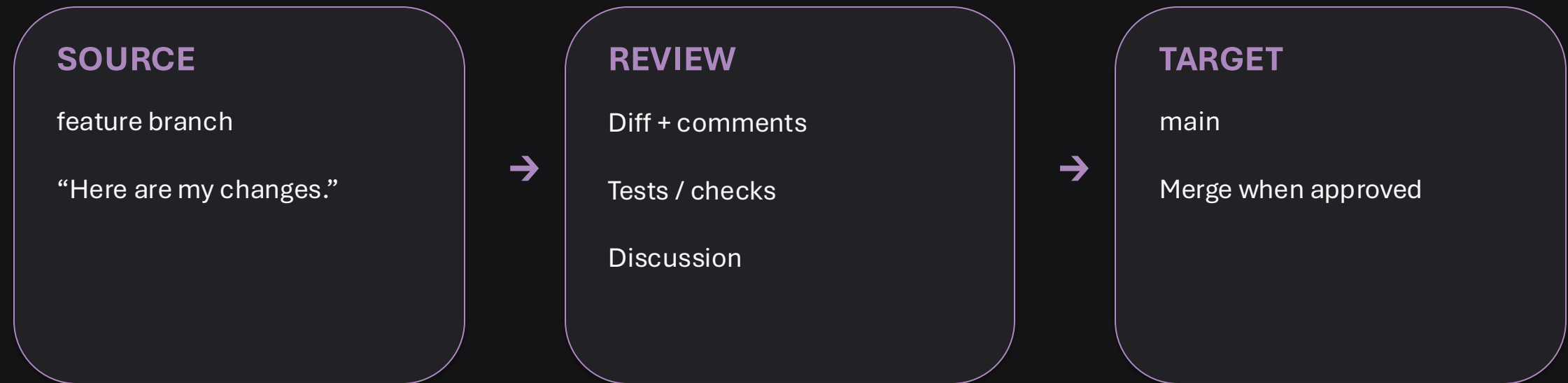
```
git push -u origin add-about-page
```

After your first push:

```
git push
```

Pull requests: Propose a change

A pull request (PR) is a proposal to merge changes from one branch into another.



Key idea: a PR is not just “uploading code”, it creates a reviewable conversation around a change.

What if something goes wrong?

“I changed the wrong file.”

git status

Review your changes before committing.

“My branch is behind.”

Update your branch from main before merging.

“Merge conflict!”

Two changes overlap.

Git needs you to choose the final version.

“I messed up.”

Don't panic.

Git history gives you checkpoints.

Stop, read the output, and ask for help if you're unsure.

Beginner Tip: read the output before running another command.

Hands-on challenge → your turn!

Make your first contribution to the WiCSE repository

1. Fork the workshop repository

2. Clone your fork locally

3. Create a branch:
feature/your-name

4. Create your profile

Add:

- Name
- Major
- Year
- Something you're learning
- GitHub username

Save it as:

- participants/your-name.md

5. Commit your changes

6. Push your branch

7. Open a pull request into main

8. Review your diff

9. Merge your contribution

SUCCESS CHECK

You should be able to explain:

fork → clone → branch → commit →
push → PR → merge

What's Next?

- Issues — track bugs, tasks, and feature requests
- Projects — organize work and project planning
- Actions — automate tests, builds, deployments, and workflows
- Forks — make your own copy of someone else's repository
- Code review — collaborate directly on proposed changes

Don't try to learn everything today. Get comfortable with the core workflow first.

Recap

GitHub is where the project lives and where people collaborate.

FORK

Make your own
copy

BRANCH

safe workspace

COMMIT

snapshot

PUSH

send to GitHub

PR

review

MERGE

integrate

If you remember one workflow, remember this.

JPMG Event



Talk Tech w/ JPMorgan Chase:
- Learn about their Software Engineering Program (SEP), network with the team, and meet panelists straight from JPMC. Put your skills to the test in a fun AI Prompt Challenge, and enjoy free JPMC swag and food throughout the event!

- Don't miss this chance to connect with industry professionals and learn more about opportunities at JPMorgan Chase.



WICSE Fall Events

SEP 10 Intro to GitHub Workshop

SEP 15 Talk Tech w/ JPMorgan Chase

SEP 17 Honeywell Ask-Me-Anything

SEP 24 Pitch Perfect & Career Ready

SEP 28 Arthrex Info Session

OCT 29 Halloween Social & Costume Contest

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stay updated!



Honeywell Event



Honeywell Ask-Me-Anything:
- This event focuses on resume building and career-fair readiness, with one-on-one resume reviews and tips to help you stand out. Come with your questions — whether it's about landing an internship, navigating the career fair, or breaking into the industry, our Honeywell guests are here to help.

- Bring your resume, bring your questions, and get ready for an open, casual conversation with real professionals in the field!